

Some review questions - 5th week

Q1 Write the 3×3 elementary matrix E_{R_1, R_2} that interchanges rows 1 and 2 of a $3 \times k$ matrix when multiplied on the left (if $A = \begin{bmatrix} a_{(1)}^T \\ a_{(2)}^T \\ a_{(3)}^T \end{bmatrix} \in \mathbb{R}^{3 \times k}$ then: $E_{R_1, R_2} A = \begin{bmatrix} a_{(2)}^T \\ a_{(1)}^T \\ a_{(3)}^T \end{bmatrix}$).

Q2 Write the 3×3 elementary matrix $E_{2R_2+R_3}$ that adds twice the second row to the third row of a $3 \times k$ matrix when multiplied on the left (if $A = \begin{bmatrix} a_{(1)}^T \\ a_{(2)}^T \\ a_{(3)}^T \end{bmatrix} \in \mathbb{R}^{3 \times k}$ then: $E_{2R_2+R_3} A = \begin{bmatrix} a_{(1)}^T \\ a_{(2)}^T \\ a_{(3)}^T + 2a_{(2)}^T \end{bmatrix}$).

Q3 Write the 3×3 elementary matrix E_{2R_2} that multiplies the second row of a $3 \times k$ matrix by 2 when multiplied on the left (if $A = \begin{bmatrix} a_{(1)}^T \\ a_{(2)}^T \\ a_{(3)}^T \end{bmatrix} \in \mathbb{R}^{3 \times k}$, then: $E_{2R_2} A = \begin{bmatrix} a_{(1)}^T \\ 2a_{(2)}^T \\ a_{(3)}^T \end{bmatrix}$).

Q4 Given $i, j \in \{1, \dots, n\}$, let $E_{R_i, R_j} \in \mathbb{R}^{n \times n}$ denote the elementary matrix corresponding to the row operation that interchanges Row i and Row j . Describe its general matrix form and express its inverse.

Q5 Given $i, j \in \{1, \dots, n\}$, let $E_{\lambda R_i + R_j} \in \mathbb{R}^{n \times n}$ denote the elementary matrix corresponding to the row operation that adds λ times Row i to Row j . Describe its general matrix form and express its inverse.

Q6 Given $i \in \{1, \dots, n\}$ and $\lambda \neq 0$, let $E_{\lambda R_i} \in \mathbb{R}^{n \times n}$ denote the elementary matrix corresponding to the row operation that multiplies Row i by λ . Describe its general matrix form and express its inverse.

Q7 Give two examples of elementary matrices of different types that are lower triangular.

Q8 Are all elementary matrices triangular? square? invertible? If the answer is no, give a counterexample.

Q9 Define the “LU decomposition”.

Q10 Explain how to obtain an LU decomposition and under what conditions it exists.

Q11 Given two square matrices A, M , prove that if M is invertible and $MA = I_n$, then A is invertible and $A^{-1} = M$.

Q12 Using the lemma “ M invertible and $MA = I_n$ implies A is invertible and $A^{-1} = M$ ”, prove that if Gauss-Jordan elimination reduces A to the identity matrix, then A is invertible.

Q13 Answer the following questions with “True” or “False”:

- If A is invertible, then $Ax = 0$ has exactly one solution.
- If $Ax = 0$ has infinitely many solutions, then A is not invertible.
- If A is not invertible, then $Ax = 0$ has more than one solution.

Q14 Answer the following questions with “True” or “False”:

- If an $n \times n$ matrix A is invertible, then there are n pivots in Gauss-Jordan elimination for solving $Ax = 0$.
- Given an $n \times n$ matrix A , if there are less than n pivots in Gauss-Jordan elimination for solving $Ax = 0$, then A is invertible.
- If an $n \times n$ matrix A is not invertible, then there are less than n pivots in Gauss-Jordan elimination for solving $Ax = 0$.

Q15 Provide two characterizations of invertibility (one involving the notion of pivots and another involving the notion of systems of linear equations).

Q16 Provide a step-by-step algorithm inspired by Gauss-Jordan elimination to compute (when possible) the inverse of a square matrix $A \in \mathbb{R}^{n \times n}$. (There could be multiple methods; just write one method is enough for this problem.)

Q17 Find the inverse of the matrix $A = \begin{bmatrix} 1 & 4 & 3 \\ -1 & -2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$.